Florian Pothin

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Quantitative Economist – Climate, Trade & Macro – Ph.D. Candidate in Economics – Université de Rennes – CREM UMR6211

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Defence expected: November 2026 – available from 2027

Profile

I study how the economy and the climate shape each other, with quantitative methods: econometrics, machine learning and structural models. A large part of the work is about measurement, because the carbon numbers this field relies on are partly chosen: my research measures how far those choices move reported emissions, how they travel from firms through trade into national statistics, and what they mean for climate policy built on reported data. Developed through a CIFRE partnership with Toovalu, a company building carbon-accounting software, where I now work as an independent consultant. Every paper ships with a public replication package.

Education

Ph.D. in Economics — *Université de Rennes – CREM UMR6211* — 2023 – Nov. 2026 (expected)

"The Decarbonization Gap: Firms, Nations, and the Economics of an Unfinished Transition"

Supervisors: Fabien Rondeau & Tovonony Razafindrabe | CIFRE industrial partnership: Toovalu (ANRT no. 2022/1627)

M.Sc. Financial and Economic Engineering — *Université de Rennes* — 2018 – 2020

Graduated with honors. Focus: macroeconomics, econometrics, data science, financial modelling, risk management.

B.Sc. Economics — *Université de La Réunion* — 2015 – 2018

Valedictorian. Focus: macroeconomics, microeconomics, mathematics, statistics, data science.

Professional Experience

Data Scientist – Research & Impact — *Toovalu, Nantes (CIFRE, now independent consultant)* — Apr. 2022 – Present

- Built econometric, forecasting and machine-learning models, with their data pipelines, embedded in a B2B SaaS platform for corporate climate strategy, ESG reporting and GHG-emissions analytics.
- Developed a firm-, sector- and country-level emissions observatory integrating CDP, ADEME, Bloomberg, EXIOBASE and macro-environmental indicators; benchmarking runs on corrected rather than reported numbers.
- Applied-AI lead: designed and shipped the platform's production LLM systems (retrieval-augmented generation, structured extraction) and its life-cycle-assessment module; built models for emissions estimation and reporting-bias correction.

Quantitative Research Intern — *Natixis Investment Managers, Paris* — Mar. 2020 – Jan. 2021

- Return-prediction models for a multi-asset fund (Python, R); ESG integration in portfolio construction and performance attribution.

Research Assistant – Macroeconomic & Country Analysis — *Université de Rennes – CREM* — Apr. – Aug. 2019

- Country-risk and macroeconomic analyses for OPEC member states; Bloomberg and World Bank data; applied research reports.

Research

Journal names are given once papers are accepted. Replication packages are public; drafts on request.

- **Voluntary Climate Pledges Widen What Firms Measure, Not What They Abate [Job Market Paper]**
  Sole-authored. Under review. Replication: 10.5281/zenodo.21627772.
  After a voluntary pledge, firms report 2.5 more Scope-3 categories within four years and reported Scope 3 rises 0.3–0.5 log points through the wider boundary alone; the rise disappears once each firm's boundary is held fixed. Scope 1 shows a precisely estimated null.
- **Beyond Reported Emissions: Uncovering Hidden Corporate Carbon Footprints**
  With F. Rondeau, T. Razafindrabe and A. Macherey. Under review, second round; revision submitted July 2026.
  Diagnoses omission and under-reporting for each of the 15 GHG-Protocol Scope-3 categories, then estimates the missing part: value-weighted, reported footprints miss 21.7% [19.5, 24.2] of the corrected total, two thirds of it in use of sold products.
- **A Product-Level Trade-LCA Framework for Embodied Carbon Accounting in Global Trade**
  Sole-authored. Under review since June 2026. Preprint: Research Square rs-10627706; code: 10.5281/zenodo.20719018.
  Links bilateral HS6 trade to life-cycle emission factors (about 5,000 products, 120 countries, 1995–2023), with LLM-assisted matching audited blind twice (17 then 8 wrong per 100). World embodied carbon in traded goods: 14,368 $MtCO_2e$ in 2023; intensity varies thirteen-fold within the median sector.
- **Transparency and Verification in the Economics of Decarbonization**
  Sole-authored. Under review. Replication: 10.5281/zenodo.21719291.
  Codes 2,578 studies on eight research fronts: 6.5% of disclosure studies analyse Scope 3, two thirds of embodied-carbon studies do not name their input–output database, and 95.7% of decoupling studies never assess Paris compliance.
- **Trade Shocks Redraw the Global Carbon Map but Do Not Shrink It**
  Sole-authored. Ready for submission. Replication: 10.5281/zenodo.21889661.
  Product-level embodied carbon through China's WTO accession, the US–China tariff war and the 2022 sanctions on Russia, generalised over 35 accessions and 9 sanction episodes: integration pulls carbon in slowly (+31% at ten years), sanctions push it out fast (up to −88%), and the carbon relocates rather than shrinking.
- **Is the Environmental Kuznets Curve Identified?**
  Sole-authored. Submission in preparation.
  A turning point counts as identified only if its bootstrap interval fits strictly inside the observed income support. On 105

countries (1995–2023), no specification passes for production- or consumption-based emissions, at 99–100% simulated power; the trade-adjusted measure passes at the threshold of the criterion.

- **Sectoral Climate Legislation and Corporate Targets**
  Sole-authored. Thesis chapter; not currently under submission.
  On 1,958 listed firms in 28 countries (2015–2023), each additional sectoral climate law raises the probability of an SBTi commitment by 1.66 percentage points; the emissions test is underpowered in available data, and the paper says so.
- **Fair Shares for Firms: What the Flat Rate Hides**
  In preparation.
  Against alternative fairness rules for the same carbon budget, the flat 4.2%-a-year rate under-charges 89% of committed Scope-1+2 tonnage, concentrated in heavy industry.
- **Endogenous Abatement as Green Capital in a Heterogeneous-Firm General Equilibrium**
  With T. Razafindrabe. In preparation.
  Hopenhayn-style general equilibrium where abatement is the green share of capital: of a 42% emission cut, 86% is reallocation across firms and 14% within-firm greening; permit design decides whether the price bites.

Funded Projects & Service

- **ANR I-DAVE** *(2024–2027)* – AI and knowledge-based framework for life-cycle-assessment reliability, with Centrale Nantes (LS2N), UTC (Roberval), Toovalu and Cetim. Grant held by Toovalu; I carry out its work on the company's side.
- **IRIS-E**, Université de Rennes – appointed member of the axis-2 working group ("Interdisciplinary and co-constructed research"), two-year mandate from December 2023.
- **SBTi Expert Working Group** – member, Data Quality Assurance and Claims, Science Based Targets initiative.

Distinctions

- **Prix Transition**, Trophées Valorisation 2024, Campus Innovation, Université de Rennes, with T. Razafindrabe and F. Rondeau, for the CIFRE CREM–Toovalu project.
- **Winner, "Prédiction de sinistres" data challenge** (Generali), 2018 edition, with M. Riou; presented at the data science seminar of Stéphane Mallat's chair, Collège de France.

Conferences & Seminars

- **International Economies: Challenges Ahead**, Ryukoku University / Yokohama National University / Sciences Po Aix / Aix-Marseille School of Economics, Kyoto (Feb.–Mar. 2026) [presentation]
- **Research workshop**, Shiga University, Japan (2026) [presentation]
- **11th International Macroeconomics Workshop**: International Macroeconomic Issues of Climate Change, CREM, Université de Rennes (2025) [presentation]
- **SESTEF 2024**, Paris (Dec. 2024) [presentation]
- **FFM29**, 29th Forecasting Financial Markets Conference, IPAG, Oxford (Sep. 2024) [presentation]

Technical Skills

Programming	Python, R, Matlab, Dynare, SQL
Econometrics & Macro	Panel data, event studies and staggered designs, randomization inference, time series (VAR/SVAR), GMM, IV/2SLS, hurdle models; heterogeneous-agent GE modelling, calibration, forecasting
Machine & Deep Learning	XGBoost, Random Forest, Elastic-Net, meta-learning; PyTorch, TensorFlow; GNN (PyTorch Geometric), CNN, RNN; LLM-assisted measurement with audited reliability
Data	Bloomberg, CDP, ADEME, EXIOBASE, BACI, World Bank, OECD.Stat; API integration, data pipelines, multi-source harmonization
Practices	Reproducible research, public replication packages, Git, HPC, LaTeX

Teaching Experience

Université de Rennes, M.Sc. Ingénierie Économique et Financière (IEF) 2023 – 2025 | ≈100h ETD
Macrodynamics: DSGE and real business cycle models | Macroeconometrics: time series

Business schools 2022 – 2026 | ≈150h total
Supervised Machine Learning in R, IA School M.Sc. (2024–2026); Python & AI Frameworks, EPSI (2022–2025); Mathematics for Finance, IGEFI (2024–2026); Green Finance, IET (2022–2023).

Languages & Interests

French (native) | English (professional working proficiency; research and teaching) | German (A2) | Japanese (A1)
Interests: macroeconomics, financial markets, ecological transition, AI for economic intelligence, open data, hiking, music.

References

Available upon request: **Fabien Rondeau** (thesis supervisor, Université de Rennes – CREM) | **Tovonony Razafindrabe** (co-supervisor, Université de Rennes – CREM)